TERMINI · AI BUILDER CUP 2026 · FUTURE OF WORK & ENTERPRISE PRODUCTIVITY

The dentist's server room

Every organisation should be able to run its own intelligence. **Termini is the layer that makes that possible** — bare hardware in, a governed AI fleet out, and the data never leaves the building.

“Your hardware. Your data. Our operations.”

Day-0 · registered in 150 ms

Day-2 · self-healing, audited

Fail-closed governance

01 A ROOM IN A CLINIC

There is a data centre behind the dental chair

A dentist in a small Korean clinic — a biology graduate, not a computer scientist — spent six months building one in his back room. He started by asking ChatGPT what a dental clinic could do with AI. He had exactly one rule.

> **Patient information does not leave this building. Not one character.**
>
> The principle the whole build was organised around

Accelerators	**12 DGX Spark-class, wired in rings of four**
Voice & documents	**2× Mac mini M5 Pro + an M4 mini**
Internal services	**3 virtualisation servers**
Power	**6 UPS units**

Telephony	**A PBX he built himself**
Models in production	**DeepSeek · GLM · others, side by side**
Data sent off-site	**None**
Who runs it	**Him, after hours**

Source: arca.live/b/aihomelab/185810092 — “A neighbourhood dental clinic's AX transformation” (13 Oct 2026; 330k+ views).

He succeeded. That is the scandal.

+ What the clinic got

- Phones answered by AI — **first token in 0.1 s**, transcribed live, summarised with a to-do list
- Insurance claims reconciled line by line — **the task staff hated most**
- Three models write a monthly diagnosis of the same numbers; **he only looks when they disagree**
- Accountant cancelled. SaaS subscriptions cancelled. Marketing spend: **zero**

− What it cost

- **Six months of evenings**, on top of dentistry
- A five-figure hardware bet, justified as **“cheaper than the salary of a bad hire”**
- Skills no clinic should have to hire for: **networking, virtualisation, PBX, inference serving**
- A single point of failure called **“the owner”**

A market that requires heroics is not yet a market. **The technology works. The product doesn't exist.**

Capability is compounding. Deployment is flat.

The most capable technology in history is not compounding where work actually happens — in the twelve-person firm, the clinic, the workshop, the practice.

95%

of enterprise GenAI pilots produced **no measurable P&L impact**

MIT NANDA, The GenAI Divide, Jul 2025 — preliminary, contested

$30–40B

of enterprise GenAI spend behind that number; **~5% reached production**

Same study — "high adoption, low transformation"

88% · 6%

of organisations use AI somewhere; **6% are high performers**

McKinsey, The State of AI, 2025

The gap is not model quality. The gap is **where the work is** — and who is allowed to put the model next to it.

Three gates between a model and the work

1 Permission

The data that would make AI valuable is the data you are **not allowed to move**: patient records, payroll, insurance claims, designs, source code.

2 Operations

On-prem means **you are the SRE**. A twelve-person firm has nobody to watch disks, memory and inference health at 3am — so it doesn't buy the box.

3 Context

Generic tools don't retain feedback or learn your workflow. The data that would teach them is **the data that can't leave**.

Three gates, one cause: **value scales with context, and context is exactly what cannot be uploaded.**

05 THE DEADLOCK

So the loop closes on itself

Entity holds the data → **It can't be sent to a cloud** → **The model stays generic** → **Worth less to that entity** → **No adoption**

↺ **AND THE CYCLE HOLDS** — BECAUSE THE NEXT BEST OPTION IS STILL TO SEND THE DATA

Cloud AI asks you to **export the asset** in order to **rent the tool**. The dentist's six months is what that bargain costs when one person refuses to accept it.

SaaS ops already proved delegation

Datadog runs best-in-class day-2 operations with an agentic surface — but it is SaaS-architected, with no self-hosted or air-gapped AI offering. It assumes the infrastructure already exists.

The top of the market is buying the boundary

Samsung × Mistral (Sept 2026) keeps AI on-prem for data sovereignty — bespoke, equity-backed, and covering models and platform, **not day-2 operations**.

Even the largest buyers are paying for the boundary. Nobody is selling it to the long tail.

So the model gets smarter every quarter, and the firm that needs it most **gets no closer to using it.**

Intelligence should be plural

Today: one model, everyone's data

One answer for everyone · value accrues to the **model owner** · shared blind spots · the entity's own context is the raw material, and it is spent.

Inverted: one model per entity

Its own data, its own answer · value accrues to the **entity** · different priors, so disagreement becomes **signal** · the context compounds instead of leaking.

More useful

A 4B model that has read twelve years of your claims beats a frontier model that hasn't. **The marginal token of general intelligence is worth less than your context in the loop.**

More reliable

Centralisation correlates failure: same weights, same blind spots. A population of sovereign models is a **portfolio**, and a jury beats an oracle.

More shareable

Evals, weights and lessons can travel between entities. **Records don't have to.** That is what enriching the ecosystem actually means.

He didn't buy intelligence. He grew one.

Three things he built, almost by accident, that we think are the product category:

A A jury

DeepSeek, GLM and others each diagnose the same month's numbers. When all three agree, it is probably real. **When they disagree, he looks.**

B An eval loop

He grades the AI's call summaries by hand — a **human-judgement screen**. That becomes ground truth, then a model comparison, then training material.

C A private corpus

Twelve years of claims, calls and outcomes that **no foundation model has ever seen** — and that no cloud vendor could be given.

Local governance is not a compliance tax. **It is the only way the feedback loop closes.**

A single case, cited as a case — not as a market.

“Isn't local just worse?”

“Frontier models keep improving. A 4B model on your box is a toy.”	The model is a **swappable part**. The product is the loop, the boundary and the operations. Swap the weights when better ones ship — Gemma today, something else next quarter.
“SMBs don't want to own infrastructure. That's why cloud won.”	Correct — which is why **we don't sell ownership**. We sell the operations that make ownership survivable: we run what they bought.
“Local AI is a compliance story, not a value story.”	**Compliance gets the budget. The eval loop keeps it.** A model graded on your own work, over your own data, is the only one that gets better at your job.

Our concession: if your work is generic, use the cloud. This is for the work that isn't — and most of the economy is work that isn't.

Six months, compressed into an afternoon

Bare node → Pre-baked image → Registered 150 ms → Plain-English intent → Guardrail clamps → Agents running

	THE DENTIST'S PATH	TERMINI'S PATH
Hardware	Wiring, hypervisor, cluster join, PBX — by hand, at night	Boot the image; the controller registers it and joins the cluster
Model	Pick, quantise, serve, benchmark, re-pick	A config knob — Gemma resident, Gemini where the customer permits it
Operations	Watch it yourself, forever	A governed operator plus a human dashboard
Governance	A rule enforced by discipline	A gate that rejects C2 fail-closed, before storage

What took him six months of evenings is **the product's first hour**.

The part the dentist had to become

Telemetry → **Threshold rule fires** → **Model decides — allowlist only** → **Audited LXD execution** → **Audit row**

1 Detect deterministically

A threshold rule fires. **Never a model.** Rules are the detector; the model is only ever the decider — so the loop works even when the model doesn't.

2 Decide inside a cage

Grammar-constrained decoding: an off-allowlist action has **no valid encoding**. Plus parameter guards, a confidence gate, and a mandatory deterministic fallback.

3 Prove it afterwards

Every autonomous change links **trigger → decision → action → result** in one audit record. “The AI did it” becomes a row you can read.

Ownership without the ops burden — **the only version of on-prem a twelve-person firm can survive.**

Two motions, one control plane

A The SME buys an outcome

Through hardware partners and integrators: a box, an afternoon, a governed fleet. **We take the operations subscription** — the part they cannot staff.

B The hardware owner licenses the layer

Colocation, edge and private-cloud operators already own the metal and the power. **We make it sellable as AI** without them building an ops team.

C The model stays a knob

Gemma by default, Gemini where the customer permits it, a different open-weight family next quarter. **No lock-in to weights** — the layer is the product.

We are not in the hardware business. **We are in the business of making other people's hardware useful.**

12 PROOF

What runs today, measured

150 ms

Node registration from the pre-baked image — **zero installs at boot**

128→8

A 128-vCPU request **clamped** to the host ceiling, with reasons

422

C2 field **rejected before storage** — fail-closed, three ways checked

1

Remediation **decided by the model**, executed, audited end-to-end

1.8 s

Diagnosis warm on the managed path — **402 s** on local CPU weights, stated not hidden

live

On GCE + Firebase + Vertex AI — **a judge can open it and watch**

Three claims, three measurements. **Nothing on this slide is a projection.**

The architecture is the governance claim

Node on premises → **On-node redactor** → **Outbound-only mTLS** → **Controller gate** → **Operator loop** → **Audit chain**

Substrate and channel — adopted, not invented

Ubuntu + an **LXD cluster** across nodes; one **outbound-only mTLS WebSocket** per node carrying heartbeat, telemetry, commands and acks. There is no inbound path from the internet to a node, and no route from an agent VM to the controller.

Boundary and loop — the two things we build

A **fail-closed** telemetry gate that rejects a non-conforming batch (422) rather than dropping fields, and a **governed operator loop**: detect → decide inside an allowlist → execute against LXD → write the audit row.

Bootstrap

Pre-baked image, single-use enrolment token, CSR → client cert, role assigned, cluster joined. **150 ms.**

Placement

Natural language → structured allocation → **guardrail clamp** → real LXD instances, rolled back on failure.

Access

A controller-held **SSH CA**: browser-generated keypair, ≤8h principal-bound certificate. No shared keys.

The node keeps the data. The controller keeps the proof. That is the whole design.

What leaves the building, exactly

CLASS	LEAVES?	CONTENTS
C0	Yes — allowlisted	Hardware and OS health: CPU, RAM, disk, GPU, inference liveness, instance inventory, rule metrics
C1	Yes — transformed	Salted hashes, bounded display labels, timestamps. Never raw identifiers
C2	**Never**	Workload payloads, call transcripts, patient or customer records, prompts, credentials, logs

The one external call — disclosed

The operator's **diagnosis** runs on Gemini via Vertex AI from the controller's loopback listener. Prompt: **C0 telemetry only**. The audit row records `egress=external`. One variable, `backend=local`, puts the same decision back on the node's own weights.

Why we publish the exception

Because a boundary you cannot see is not a boundary. The claim is not "nothing leaves" — it is **"nothing C2 leaves, and everything that does is named, classified and audited."**

Gemini decides. Gemma stays home.

1 Gemini 2.5 Flash · Vertex AI

The **diagnosis** — naming the remediation from C0 telemetry. Measured 1.8 s warm. The only call that leaves the premises, and it says so in the audit trail.

2 Gemma 3 4B · on Node C

Google's **open weights**, served by `llama.cpp` on CPU-only GCE. Agents and intent extraction run here and **forward nothing**. 8–10 tok/s — stated, not hidden.

3 Cloud Run · GPU reference

The same Gemma on **1× L4** with vLLM, deployed as the managed reference model — and the shape this takes when GPU quota stops being the blocker.

GCE · controller · **GCE · node C** · **Firebase Hosting · dashboard** · **Cloud Storage · weights** · **Pre-baked node image**

Open weights on the customer's metal. A managed model where the customer permits it. **The same code path either way.**

From one clinic to a fleet of them

Real bare-metal onboarding

MAAS + iPXE netboot, so a box arrives from a partner and joins without a technician on site.

A GPU node in the demo

L4 quota is the current blocker; the path is already built and served on Cloud Run.

Learned baselines

Edge anomaly detection replacing static thresholds — the operator learns each site's normal.

Federated eval exchange

Evals and weights travel between entities; records don't. The plural-ecosystem mechanism, productised.

Compliance mapping

GDPR, EU AI Act, Korea PIPA, India DPDP posture mapped onto the C0/C1/C2 boundary.

Hardware partner channel

The box is commodity. The operator is the product. Partners sell metal; we run it.

The long tail does not need a data centre. **It needs one box, one afternoon, and someone else on call.**

TERMINI · AI-INFRA

He built this alone. Nobody should have to.

Bare hardware in. A governed AI fleet out. Your data stays where it is — and somebody else is on call at 3am.

“Your hardware. Your data. Our operations.”

Live demo | Architecture | Questions